

DRPO Pipeline v2.3 Core: Faithful Outline-to-Blueprint Slice

Anonymous review artifact

Scope. This two-page artifact validates the manuscript pipeline, not a new experiment. It preserves the approved paragraph identities `METHOD-P03` and `EXP-P04` without merge, split, rename, or reorder. It uses the long-run-validated C-U1-E3-ADAM-RERUN compact repository evidence. C-U1 evaluates same-distribution held-out-context generalization, and task, boundary, and NaN/Inf events remain separate.

Proposition 2: Vanishing Weighted Far-Field Gradient

The exponential envelope is selected for a precise tail property rather than an assumed utility-decay law. Under the finite-order score-growth condition in Proposition 2, multiplying the unweighted score-times-advantage contribution by $\exp(-\lambda r)$ drives the weighted far-field term to zero as policy-relative remoteness grows. This is a method-level tail guarantee: it neither ranks all tapers nor claims that sample utility itself decays exponentially.

Proposition 2 (Vanishing weighted far-field gradient). *Let $r \geq 0$ be policy-relative remoteness and suppose the unweighted score-times-advantage contribution satisfies $\|g(r)\| \leq C(1+r)^k$ for finite constants $C > 0$ and $k \geq 0$. For any $\lambda > 0$, define $\tilde{g}(r) = e^{-\lambda r}g(r)$. Then $\|\tilde{g}(r)\| \rightarrow 0$ as $r \rightarrow \infty$.*

Proof. The assumption gives

$$\|\tilde{g}(r)\| \leq C(1+r)^k e^{-\lambda r}.$$

For integer $m > k$, the exponential series implies $e^{\lambda r} \geq (\lambda r)^m / m!$ for $r > 0$. Hence

$$(1+r)^k e^{-\lambda r} \leq \frac{m!(1+r)^k}{(\lambda r)^m},$$

and the right-hand side converges to zero because $m > k$. The squeeze theorem proves the result. The proposition controls the weighted gradient tail; it makes no assumption that sample utility decays exponentially. $\square$

Claim boundary. The proposition controls the weighted gradient tail under finite-order score growth. It does not assume exponential utility decay and does not establish a universal ranking among taper families.

RQ2b: Targeted Causal Transmission

Table 1: Terminal outcomes. Task, support-boundary, and NaN/Inf events are separate.

Method	Fixed reward (95% CI)	Task	Support	NaN/Inf
Baseline	2.22e-06 [2.10e-06, 2.35e-06]	20/20	20/20	0/20
Near-zero	2.18e-06 [2.05e-06, 2.30e-06]	20/20	20/20	0/20
Far-zero	0.739 [0.739, 0.740]	0/20	0/20	0/20
Far-cap	0.733 [0.733, 0.734]	0/20	0/20	0/20
Global-scale	0.599 [0.599, 0.599]	0/20	0/20	0/20
Far-to-near	0.875 [0.875, 0.876]	0/20	–	0/20

We test the registered transmission path with four matched interventions over 20 paired held-out seeds and report task, boundary, and numerical events separately. In the fixed-variance branch, Baseline and Near-zero finish at rewards 2.22e-06 and 2.18e-06 and undergo task-performance collapse in 20/20 and 20/20 seeds. Far-zero instead reaches 0.739 [0.739, 0.740], and Far-cap reaches 0.733 [0.733, 0.734], with no task collapse. The registered budget controls also remain non-collapsed: Global-scale reaches 0.599, while transferring the far budget to the near component reaches 0.875; these controls diagnose the pathway rather than establish a universal method ranking. The learnable-variance branch reaches the support/variance-contraction boundary in 20/20 Baseline and 20/20 Near-zero seeds, with mean onsets at steps 72.9 and 73.1, whereas Far-zero and Far-cap record 0/20 and 0/20 such events. All four methods retain finite parameters and record 0/20 NaN/Inf failures. Within this controlled same-distribution held-out-context setting, the near-removal non-rescue, far removal/capping rescue, and fixed-budget controls identify the far-field component as the dominant causal transmission path without extending that conclusion to universal off-policy behavior.

Pipeline audit

The approved outline is compiled into a 39-node AST. Every node receives an explicit enabled or disabled status; the two enabled nodes map one-to-one into the blueprint and prose. Every empirical number is rendered from `research_snapshot.json`. Missing evidence fails closed rather than creating a placeholder result.

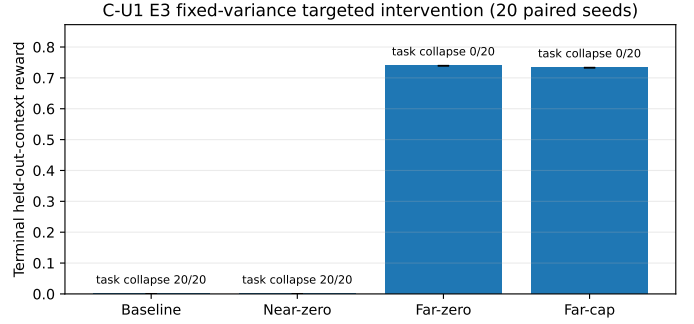


Figure 1: Fixed-variance terminal reward with 95% CIs; labels show task-collapse counts.